

Detection, Localization and Tracking of Underwater Acoustic Sources via A Passive Fiber Optic Hydrophone Array

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Abstract A fiber optic hydrophone array comprising four passive fiber mandrels, interrogated by a distributed acoustic sensing system was deployed in a lake for field testing. Several stationary and moving underwater acoustic sources were localized, continuously tracked, and their acoustic signals were enhanced by beamforming techniques. ©2026 The Author(s)

Introduction

Distributed fiber optic sensing (DFOS) is a well-established sensing technology, that can transform kilometers long standard communication optical fibers into passive sensors [1]. Even though there are many different methods and realizations of DFOS systems, the prevalent ones are Distributed Temperature Sensing (DTS) [2], Distributed Vibration Sensing (DVS) [3], Distributed Strain Sensing (DSS) [4] and Distributed Acoustic Sensing (DAS) [5]. Since these methods are single ended measurements, where the interrogator is connected to only one end of the fiber cable, and do not require additional connections to the sensing region such as for supplying power or transmitting the data, the sensing system can be far away from the sensing region. This allows these systems to have very unique use-cases and applications such as in smart cities, traffic monitoring, rain detection, border security, intrusion detection, infrastructure monitoring, Earthquake monitoring and even entomological observations [6-10].

When it comes to sensing, underwater environment brings many challenges where standard solutions suffer. Present underwater monitoring systems rely on sparse, point-based sensors such as electric hydrophones and sonar platforms, which provide limited spatial coverage and require costly deployment and maintenance. A major practical challenge is supplying power to these sensors and maintaining reliable communication links, as underwater environment severely constrains wireless transmission and often require multiple cables or periodic retrieval for data access. DAS technology addresses many of these limitations by enabling continuous, wide-area monitoring along a passive communication optical fiber, eliminating the requirement for in situ power and data communication infrastructure while providing real time sensing with high spatial resolution. Some previous work relies on underwater telecommunication cables to monitor sub-sea infrastructures [11], marine life [12] and ship detection and identification [13].

In this work, we deployed a passive fiber optic hydrophone (FOH) array comprising 4 mandrel

type FOHs in Mercer Lake NJ as a real-world underwater environment for harbor security applications. As part of the experiments, we incorporated several underwater acoustic targets, including an underwater speaker deployed at two locations to generate whale calls, impulsive events from rock throws as stationary targets, and an unmanned underwater vehicle (UUV) as a moving acoustic target.

Field Deployment and Experimental Setup

Each FOH, pictured in Figure 1a, employs ~20 meter long fiber cable wrapped around a cylindrical structure to enhance the sensitivity to acoustic disturbances. The FOHs are connected in series by fusion splicing and deployed in Mercer lake by

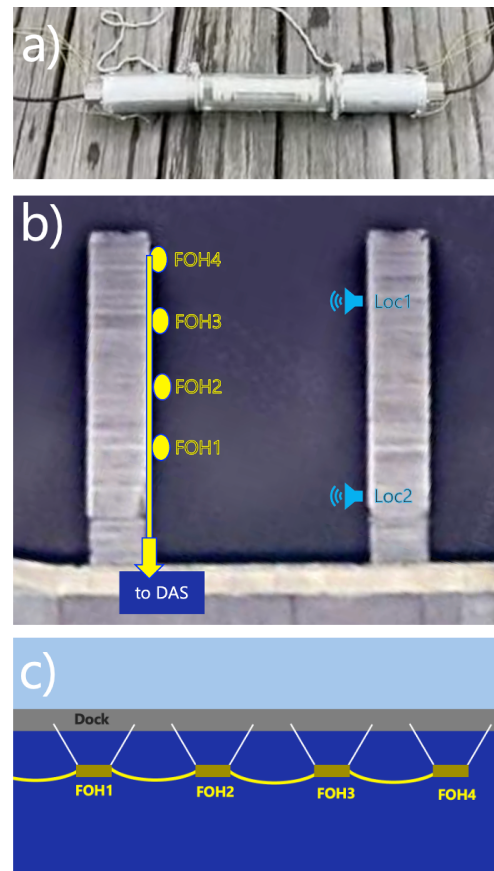


Fig. 1: a) Image of the FOH, b) top view of the deployment site, showing FOH and the underwater speaker locations and c) the side view of the FOH array deployment.

a dock in a linear array with a spacing of ~ 2.5 meters between them. The array was suspended at a depth of ~ 1 meters using ropes as shown in Figure 1b (top view) and Figure 1c (side view).

The DAS system was set at a spatial resolution of 1 meters, the pulse width was set to 40 ns, with a repetition rate of 20 kHz, the gauge length was set as 4 meters.

In this study, we have evaluated three different targets. The first target was a stationary underwater speaker at a depth of ~ 1 meter, emitting a whale call at two different locations on the nearby dock, shown in Figure 1b as Loc1 and Loc2. The second type of stationary targets was generated by randomly thrown small rocks to emulate impulsive acoustic events. Finally, the third target was a commercially available UUV, traveling in different paths at a depth of ~ 1.5 meter, controlled by a human operator. For each of the targets, the acoustic source was successfully detected, localized and in the case of moving targets, continuously tracked. In addition, their acoustic signals were recovered via beamforming to further improve the signal quality.

Localization of Static Underwater Targets

For each of the tests, the acoustic signals are captured by the 4 FOHs simultaneously, and the relative delay in time of arrival was calculated by Generalized Cross-Correlation with Phase Transform (GCC-PHAT) algorithm [14]. This algorithm estimates the time delay between two signals by computing their cross-power spectrum, normalizing it by its magnitude to retain only the phase information, and then taking the inverse FFT to obtain a sharpened cross-correlation whose peak indicates the delay. This phase-only weighting suppresses the influence of signal amplitude and improves robustness in noise and reverberation.

Once the time differences of arrival (TDOA) between each FOH pair are determined, the source location can, in principle, be calculated analytically based on geometry. However, due to potential mismatches in the relative delays, the source location may not always be analytically solvable. To address this, we instead define a search grid with a resolution of 25 centimeters over the region of interest and evaluate an error function at each candidate location. The source position is then inferred by identifying the point that minimizes this error. This way we can obtain the most likely acoustic source location as a probability distribution.

Figure 2 shows the calculated locations for a) the whale call when the underwater speaker was at location 1, and b) when it was at location 2, c) rock throw #1 and d) rock throw #2. The yellow circles represents the FOH locations and FOH1 was arbitrarily chosen as the origin.

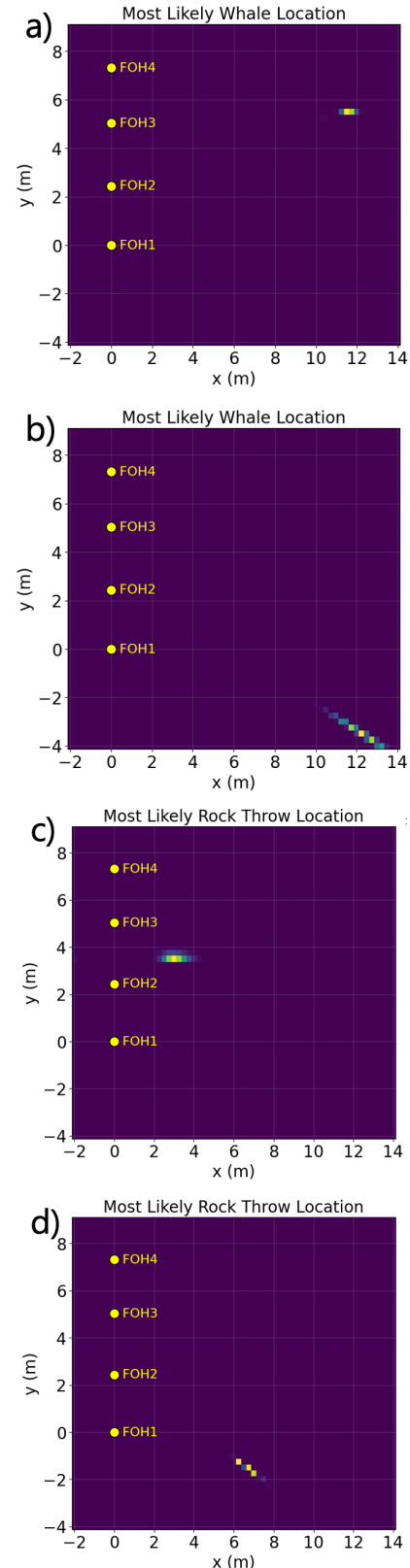


Fig. 2: The most probable locations calculated for a) the whale call sound played at location 1, b) at location 2, c) first rock throw, and d) second rock throw.

Tracking of Moving Underwater Targets

For the continuous tracking of a moving acoustic target, in our case a commercially available

unmanned underwater vehicle, the detected signals are split into 0.5 seconds long segments and the relative delays are continuously calculated for each segment, and the most probable target location is updated at every step. To visualize the trajectory of the UUV, the most likely target positions are combined into an interpolated line on a 2D plane as shown in Figure 3. The UUV was kept at a fixed depth of ~ 1.5 meter and was remotely controlled by a human operator along different trajectories and at different speeds.

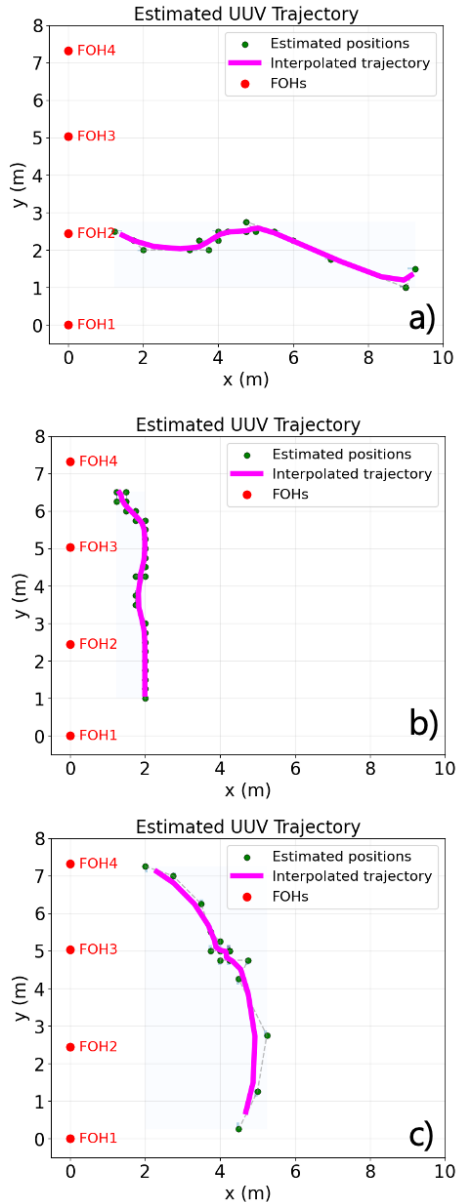


Fig. 3: Calculated trajectories of the UUV for different paths at a depth of ~ 1.5 meter.

Improved Audio Recovery

In addition to localization and tracking of underwater targets, the recorded acoustic signals were enhanced using delay-and-sum beamforming to improve the signal-to-noise ratio (SNR). Since the acoustic signals were acquired

simultaneously by four FOHs and their relative times of arrival were estimated, a delay-and-sum beamforming was performed by aligning the signals in time domain according to these delays and coherently summing them.

This process reinforces the desired signal, which adds constructively across all channels, while uncorrelated noise and interference tend to average out. As a result, the delay-and-sum beamforming improves SNR, increases robustness to noise, and provides partial suppression of off-axis interference, leading to a cleaner and more reliable recovered acoustic signal.

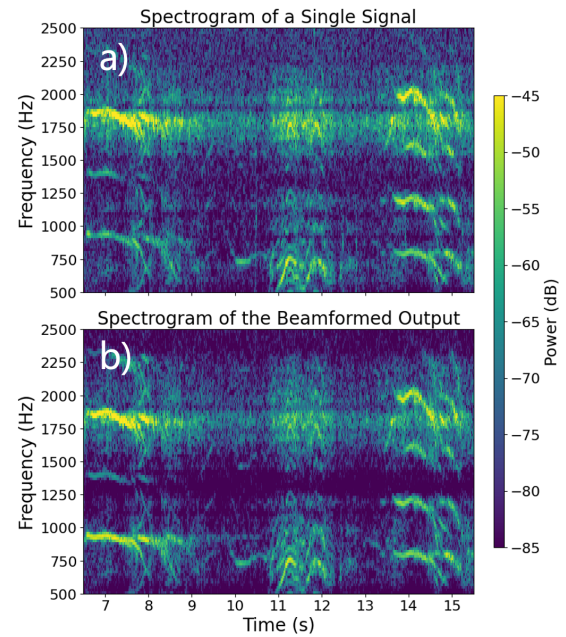


Fig. 4: The spectrogram of the recorded whale call signal: a) captured by a single FOH, and b) after delay-and-sum beamforming.

Figure 4a shows the exemplary whale call spectrogram recorded by a single FOH, while Figure 4b shows the spectrogram of the beamformed output. Signal-to-Noise Ratio calculations indicate an average improvement of 3.6 dB using delay-and-sum beamforming.

Conclusions

In this study we deployed a fiber optic hydrophone array consisting of four fiber mandrels in a linear configuration in a lake environment for harbor security applications. The FOH array was interrogated using a distributed acoustic sensing system. We successfully localized several stationary underwater acoustic targets, including an underwater speaker and rock throws, and continuously tracked a UUV moving along different trajectories. In addition, delay-and-sum beamforming method was applied to enhance the recorded acoustic signals, achieving an average signal-to-noise ratio improvement of 3.6 dB.

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